ANALYTICAL CHEMISTRY CUMULATIVE EXAM

March 7, 1992

The nitrobenzene in 210 mg of an organic mixture was reduced to phenylhydroxylamine at a constant potential of -0.96 V (vs SCE) applied to a mercury cathode. The reaction is:

$$C_6H_5NO_2 + 4H^+ + 4e^- \rightleftharpoons C_5H_5NHOH + H_2O$$

The sample was dissolved in 100 mL of methanol; after electrolysis for 30 min. the reaction was judged complete. An electronic coulometer in series with the cell indicated that the reduction required 29.74 Coulombs of charge. Calculate the percentage of $C_6H_5NO_2$ in the sample. Useful information: Faraway Constant = 96,500.

At a potential of -1.0 V (verses SCE), $CCl_4$ in methanol is reduced to $CHCl_3$ at a mercury cathode by the reaction:

$$2CCl_4 + 2H^+ + 2Hg(l) + 2e^- \rightleftharpoons 2CHCl_3 + Hg_2Cl_2(s)$$

At -1.80 V, the $CHCl_3$ further reacts to give $CH_4$ by the reaction:

$$2CHCl_3 + 6H^+ + 6Hg(l) + 6e^- \rightleftharpoons 2CH_4(g) + 3Hg_2Cl_2(s)$$

A 0.75 gram sample containing $CCl_4$, $CHCl_3$, and inert organic species was dissolved in methanol and electrolyzed at -1.0 V until the current approached zero. A coulometer indicated that 11.63 coulombs of charge was required to complete the reaction. The potential of the cathode was adjusted to -1.8 V. The current then increased again eventually returning to zero, the coulometer recording an additional 6.8 coulombs of charge. Calculate the weight percent of $CCl_4$ and $CHCl_3$ in the sample mixture.